

Customer Shopping Behavior Analysis

1. Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

2. Dataset Summary

- Rows: 3,900
- Columns: 18
- Key Features:
 - Customer demographics (Age, Gender, Location, Subscription Status)
 - Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)
 - Shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)
- Missing Data: 37 values in Review Rating column

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

- **Data Loading:** Imported the dataset using `pandas`.
- **Initial Exploration:** Used `df.info()` to check structure and `.describe()` for summary statistics.

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	1
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	1
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	22
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN

Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
3900	3900	3900.000000	3900	3900
2	2	NaN	6	7
No	No	NaN	PayPal	Every 3 Months
2223	2223	NaN	677	584
NaN	NaN	25.351538	NaN	NaN
NaN	NaN	14.447125	NaN	NaN
NaN	NaN	1.000000	NaN	NaN
NaN	NaN	13.000000	NaN	NaN
NaN	NaN	25.000000	NaN	NaN
NaN	NaN	38.000000	NaN	NaN
NaN	NaN	50.000000	NaN	NaN

- **Missing Data Handling:** Checked for null values and imputed missing values in the `Review Rating` column using the median rating of each product category.
- **Column Standardization:** Renamed columns to **snake case** for better readability and documentation.
- **Feature Engineering:**
 - Created `age_group` column by binning customer ages.
 - Created `purchase_frequency_days` column from purchase data.
- **Data Consistency Check:** Verified if `discount_applied` and `promo_code_used` were redundant; dropped `promo_code_used`.
- **Database Integration:** Connected Python script to MS_SQL and loaded the cleaned DataFrame into the database for SQL analysis.

4. Data Analysis using SQL (Business Transactions)

We performed structured analysis in MS_SQL to answer key business questions:

1. **Revenue by Gender** – Compared total revenue generated by male vs. female customers.

Results Messages		
	Gender	Total_Revenue
1	Male	157890
2	Female	75191

2. **High-Spending Discount Users** – Identified customers who used discounts but still spent above the average purchase amount.

Results Messages			
	Customer_ID	Purchase_Amount_USD	Discount_Applied
1	43	100	Yes
2	96	100	Yes
3	194	100	Yes
4	205	100	Yes
5	244	100	Yes
6	249	100	Yes
7	456	100	Yes
8	519	100	Yes
9	582	100	Yes
10	616	100	Yes
11	770	100	Yes
12	862	100	Yes
13	1209	100	Yes
14	1301	100	Yes
15	1406	100	Yes
16	1413	100	Yes
17	1422	100	Yes
18	1457	100	Yes
19	1480	100	Yes
20	1592	100	Yes
21	1567	99	Yes
22	1588	99	Yes

3. **Top 5 Products by Rating** – Found products with the highest average review ratings.

Results Messages			
	Item_Purchased	Average_Review_Rating	Total_Purchases
1	Gloves	3.860000	140
2	Sandals	3.840000	160
3	Boots	3.820000	144
4	Hat	3.800000	154
5	Skirt	3.790000	158

4. **Shipping Type Comparison** – Compared average purchase amounts between Standard and Express shipping.

Results Messages		
	Shipping_Type	Average_Purchase_Amount
1	Express	60.480000
2	Standard	58.460000

5. **Subscribers vs. Non-Subscribers** – Compared average spend and total revenue across subscription status.

Results Messages				
	Subscription_Status	Total_Customers	Average_Spend	Total_Revenue
1	No	2847	59.870000	170436
2	Yes	1053	59.490000	62645

6. **Discount-Dependent Products** – Identified 5 products with the highest percentage of discounted purchases.

Results		Messages		
	Item_Purchased	Total_Purchases	Discounted_Purchases	Discount_Rate_Percent
1	Hat	154	77	50.000000000000
2	Sneakers	145	72	49.660000000000
3	Coat	161	79	49.070000000000
4	Sweater	164	79	48.170000000000
5	Pants	171	81	47.370000000000

7. **Customer Segmentation** – Classified customers into New, Returning, and Loyal segments based on purchase history.

Results		Messages	
	Customer_Segment	Number_of_Customers	
1	Loyal	3116	
2	Returning	701	
3	New	83	

8. **Top 3 Products per Category** – Listed the most purchased products within each category.

Results		Messages		
	Category	Product_Rank	Item_Purchased	Total_Purchases
1	Accessories	1	Jewelry	171
2	Accessories	2	Belt	161
3	Accessories	3	Sunglasses	161
4	Clothing	1	Blouse	171
5	Clothing	2	Pants	171
6	Clothing	3	Shirt	169
7	Footwear	1	Sandals	160
8	Footwear	2	Shoes	150
9	Footwear	3	Sneakers	145
10	Outerwear	1	Jacket	163
11	Outerwear	2	Coat	161

9. **Repeat Buyers & Subscriptions** – Checked whether customers with >5 purchases are more likely to subscribe.

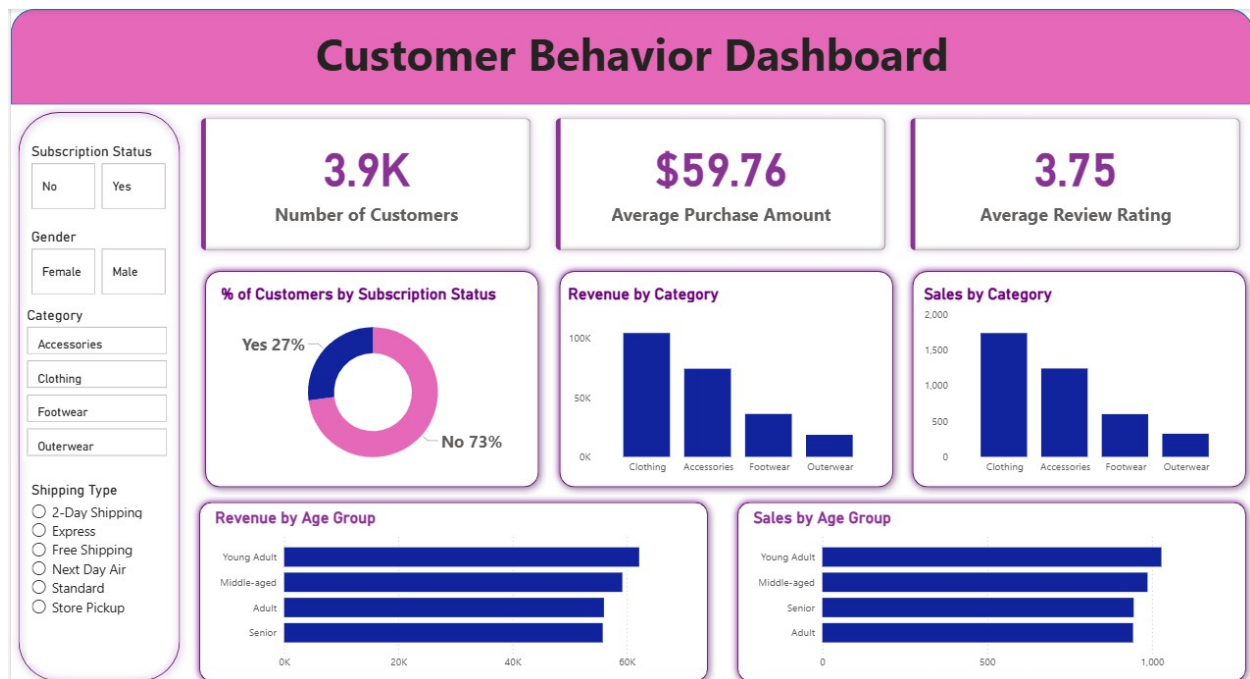
Results		Messages	
	Subscription_Status	Repeat_Buyers	
1	No	2518	
2	Yes	958	

10. **Revenue by Age Group** – Calculated total revenue contribution of each age group.

Results		Messages		
	Age_Group	Total_Customers	Total_Revenue	Revenue_Contribution_Percent
1	56+	1105	65256	28.000000000000
2	46-55	753	45619	19.570000000000
3	26-35	742	44342	19.020000000000
4	36-45	729	43234	18.550000000000
5	18-25	571	34630	14.860000000000

5. Dashboard in Power BI

Finally, we built an interactive dashboard in **Power BI** to present insights visually.



6. Business Recommendations

- **Boost Subscriptions** – Promote exclusive benefits for subscribers.

- **Customer Loyalty Programs** – Reward repeat buyers to move them into the “Loyal” segment.
- **Review Discount Policy** – Balance sales boosts with margin control.
- **Product Positioning** – Highlight top-rated and best-selling products in campaigns.
- **Targeted Marketing** – Focus efforts on high-revenue age groups and express-shipping users.